Supplementary Material: IDT-Calibrated Latent Transport for Style-ID Artistic Transfer

Anonymous Submission

Overview

This supplement consolidates the paper-facing artifact paths, protocol details, and auxiliary evaluations used by the active AAAI-27 draft. It covers four items that are only summarized in the main paper:

- the Distinct5-WikiArt split and the selected operating-point packets;
- the target-pooled ArtFID recompute path used for all main Distinct5 comparisons;
- the fixed Seedream-4.5 API protocol and repaired-750 assembly rule;
- the non-CLIP Distinct5 style probe used as an auxiliary target-style verification path.

Distinct5-WikiArt Split

The main split uses five WikiArt domains: `Early_Renaissance`, `Impressionism`, `Minimalism`, `Rococo`, and `Ukiyo_e`. The current local roots are:

- train: `Dataset/distinct5_512/train`
- test: `Dataset/distinct5_512/test`

The current paper-facing classifier probe uses all 5,000 train images and all 150 held-out test images. The evaluation packet always uses the full 5×5 protocol over 750 generated outputs unless explicitly marked as transfer-only post-aggregation.

Main-Point Ledger

The compact artifact ledger is stored in:

- `aaai2027/main_point_artifact_ledger.csv`

This file ties each main manuscript point to its config or protocol file, eval root, target-pooled ArtFID recompute artifact, and auxiliary metrics table.

Operating-Point Mechanism Table

The main paper now defines the paper-facing variants in words. Table 2 gives the compact mechanism version.

Target-Pooled ArtFID Recompute

The main paper no longer mixes the older pair-mean JSON view with the current target-pooled reading. The current fast recompute script is:

- `tools/compute_targetwise_artfid_fast.py`

Its contract is:

1. pool generated images by target style;
2. recompute the art-domain FID term on the pooled target bucket;
3. recompute the LPIPS-Alex content term on the same pooled scope;
4. form per-target ArtFID as $(1 + \mathrm{FID}_{art})(1 + \mathrm{LPIPS}_{content})$;
5. report the mean of the five per-target ArtFID values.

This reproduces the trusted Distinct5 reference packets to rounding tolerance:

- IDT all-pairs: `216.5`
- SaMST e15 all-pairs: `395.7`

The current paper-facing recompute artifacts are:

- `aaai2027/local_eval/lbm_k_e1_artfid_fast_repro.`
- `aaai2027/local_eval/lbm_knee_e13_artfid/aggrega`
- `aaai2027/local_eval/lbm_psv2_e13_artfid_fast_re`
- `aaai2027/tmp_samst_artfid_fast_repro3.json`
- `aaai2027/local_eval/seedream_repaired750_artfid`

Fixed-Rule Follow-Up Splits

Two reviewed fixed-rule follow-up WikiArt stress splits are already closed and safe to cite as narrow generalization support:

- `docs/experiments/2026-06-06-faraday-split1-pape`
- `docs/experiments/2026-06-06-faraday-split2-pape`

Closed transfer-only readout:

- split1 IDT: `0.6858`; split1 LBM-F e1: `0.6997`; gain over IDT: `+0.0139`
- split2 IDT: `0.7168`; split2 LBM-F e1: `0.7241`; gain over IDT: `+0.0073`

These packets are intentionally narrow:

Table 1: Paper-facing operating-point ledger.

Point	Role	Config or Protocol
IDT	no-op floor	`idt_5x5 summary`
LBM-K e1	conservative anchor	`distinct5_512_ema_variant_k_content_adaptive_vq_queue_e3.json`
LBM-Knee e13	main closed point	`inmortal_xpred_kmanifold_pattn_anisostokes_queue_from_pattn_seed42_b8`
LBM-PS-v2 e13	style ceiling	`inmortal_xpred_kmanifold_pattn_stokes002_from_pattn_seed42_b16.json`
SaMAM-2250	negative compact baseline	`2026-06-04-samam-distinct5-valid-through-2250.md`
SaMST e15	high-style baseline	`samst_distinct5_512_real_b2_e15_20260602/epoch_0015 summary`
Seedream-4.5	external large-prior reference	`assembly_manifest.json`

Table 2: Compact mechanism ladder for the current paper-facing LBM rows.

Variant	Endpoint selection	Texture/statistics	Motion control
LBM-K	queue-guided OMF	SA-SWD	kinetic-only
LBM-Knee	barycentric top-8 + lowfreq/edge	cross-attn proximal	anisotropic + Stokes + queue smoothing
LBM-PS	barycentric top-8 + lowfreq/edge	cross-attn proximal	Stokes 0.03
LBM-PS-v2	barycentric top-8 + lowfreq/edge	cross-attn proximal	Stokes 0.02

Table 3: Transfer-only non-CLIP target-style verification.

Point	Target Acc.	Target Prob.	Source Prob.	Margin
IDT	0.010	0.0168	0.9329	-0.9161
LBM-K	0.167	0.1524	0.7069	-0.5544
LBM-Knee	0.237	0.2123	0.5633	-0.3511
LBM-PS-v2	0.272	0.2696	0.3064	-0.0368
SaMST e15	0.248	0.2405	0.5953	-0.3548
Seedream-4.5	0.378	0.3758	0.4774	-0.1016

- they show that the unchanged-image pathology persists beyond Distinct5-WikiArt;
- they show that a compact retained LBM point still clears the split-local IDT floor;
- they do not support broader baseline or artifact-superiority claims on those follow-up splits.

Seedream-4.5 Protocol

Seedream-4.5 is treated strictly as an external large-prior reference, not as a same-interface style-ID baseline. The fixed evaluated package is:

- `Related_Works/baseline_pipeline/results/seedream45_api/distinct5_512_seedream45_windhub_202`

Protocol summary:

- provider: `windhub.cc`
- model id: `doubao-seedream-4-5-251128`
- inference interface: source image plus a fixed target-domain prompt template
- repaired assembly: the first full pass yielded 720 images; 30 blocked jobs were repaired using same-style replacement sources
- replacement map: `docs/experiments/2026-06-07-distinct5_seedream45_replacements.json`
- repaired source root for pooled content comparisons: `tmp/distinct5_seedream_repaired_eval_20260607`

The current same-scope target-pooled recompute yields:

- all-pairs ArtFID: `311.5`
- transfer-only ArtFID: `361.6`

Non-CLIP Style Probe

The non-CLIP style verification path uses the existing ConvNeXt-Tiny image classifier trainer:

- script: `src/utils/classify.py`
- output checkpoint: `aaai2027/distinct5_convnext_style_classifier.pt`
- output report: `aaai2027/distinct5_convnext_style_cla`

Training setup:

- train root: `Dataset/distinct5_512/train`
- val root: `Dataset/distinct5_512/test`
- image size: `224`
- batch size: `40`
- backbone: pretrained ConvNeXt-Tiny

Held-out test result:

- accuracy: `0.9600`
- macro recall: `0.9600`
- mean confidence: `0.9643`

The paper-facing operating-point scoring script is:

- `tools/eval_nonclip_style_probe.py`

Outputs:

- `aaai2027/distinct5_nonclip_style_probe.csv`
- `aaai2027/distinct5_nonclip_style_probe.json`

Row-Resampled IDT Bootstrap

The current paper keeps row-resampled transfer-only bootstrap as a narrow stability check rather than a full clustered uncertainty analysis. The extended artifact package is:

- `aaai2027/distinct5_idt_bootstrap_extended.csv`
- `aaai2027/distinct5_idt_bootstrap_extended.md`

For the promoted rows, all resampled draws remain above the IDT floor:

- LBM-Knee e13: $\Delta_{\mathrm{idt,tr}} = 0.0702$, interval $[0.0635, 0.0769]$
- LBM-PS-v2 e13: $\Delta_{\mathrm{idt,tr}} = 0.0901$, interval $[0.0826, 0.0976]$

- Seedream-4.5: $\Delta_{\mathrm{idt,tr}} = 0.0538$, interval $[0.0484, 0.0592]$

These are row-resampled intervals over paired transfer rows, not clustered source/style confidence intervals.

Randomness and Seeds

The current paper-facing Distinct5 packets use fixed seeds where the reviewed local configuration exposes them directly:

- the current `aaai2027` successor configs use seed 42 and encode this in the config names;
- the non-CLIP ConvNeXt style probe uses seed 42;
- the compact LBM-K anchor is referenced through its stored resolved config and eval root under the original run directory;
- Seedream-4.5 is treated as an external API reference and does not expose a user-controlled sampling seed in the current provider interface.

Reproduction Scripts

The main paper-facing scripts referenced by the current draft are:

- `tools/compute_targetwise_artfid_fast.py`
- `tools/eval_selected_style_metrics.py`
- `tools/eval_content_edge_purity.py`
- `tools/eval_nonclip_style_probe.py`
- `tools/compute_distinct5_idt_bootstrap.py`
- `tools/count_paper_point_params.py`
- `aaai2027/scripts_gen_distinct5_page1_summary.py`
- `aaai2027/scripts_gen_distinct5_qualitative_main.py`

Parameter Counts

The current paper-facing Distinct5 checkpoints can be counted with:

- `tools/count_paper_point_params.py`
- output: `aaai2027/paper_point_param_counts.csv`

Current reviewed counts:

- LBM-K: `1.982M`
- LBM-Knee: `1.946M`
- LBM-PS-v2: `1.946M`

These counts are for the current Distinct5-WikiArt packet family. They are reported separately from the preserved historical CycleGAN-256 row in the main paper.

Qualitative Provenance Note

The current main-paper qualitative strip uses two Distinct5 cases that are available across all current paper-facing families. For the SaMAM column, the source image is a crop from the existing `distinct5_visual_alignment_grid.jpg` artifact because the validated SaMAM-2250 raw image packet is remote-only in the current workspace state. The quantitative manuscript boundary remains the validated SaMAM-2250 packet.

Additional Qualitative Gallery

The supplement adds six aligned Distinct5 cases that keep the current paper-facing method set: `Source | IDT | SaMAM-2250 | SaMST | Seedream-4.5 | LBM-K | LBM-Knee | LBM-PS-v2 | target ref`. These rows are intended to show a broader spread than the two main-paper cases and to expose both conservative and failure-like operating regimes.

Blind Pairwise Packet

A blinded A/B comparison packet has been prepared for the highest-value paper comparisons:

- LBM-Knee vs SaMST
- LBM-Knee vs Seedream
- LBM-PS-v2 vs SaMST
- LBM-K vs IDT

Bundle contents:

- `aaai2027/blind_pairwise_v1/blind_pairwise_manif`
- `aaai2027/blind_pairwise_v1/blind_pairwise_rubri`
- `aaai2027/blind_pairwise_v1/exploratory_blind_au`
- `aaai2027/blind_pairwise_v1/exploratory_blind_au`
- `aaai2027/blind_pairwise_v1/exploratory_blind_au`
- `aaai2027/blind_pairwise_v1/panels/`

Each panel shows `Source | Candidate A | Candidate B | Target ref`, with randomized A/B order stored only in the manifest. This packet is ready for human or external VLM scoring. Local exploratory blind rubric notes and their summaries are kept only as internal supplementary support and are not promoted as a main paper claim surface.

Software and Hardware

Current local analysis environment:

- Python: `3.12.0`
- GPU: `NVIDIA GeForce RTX 4070 Laptop GPU`, 8 GiB VRAM
- driver: `610.47`
- package snapshot: `aaai2027/local_analysis_requirement`
- environment template: `aaai2027/local_analysis_environment.yml`

Remote training timings reported in the main paper come from the reviewed single-lane packet logs and are not recomputed by this supplement.

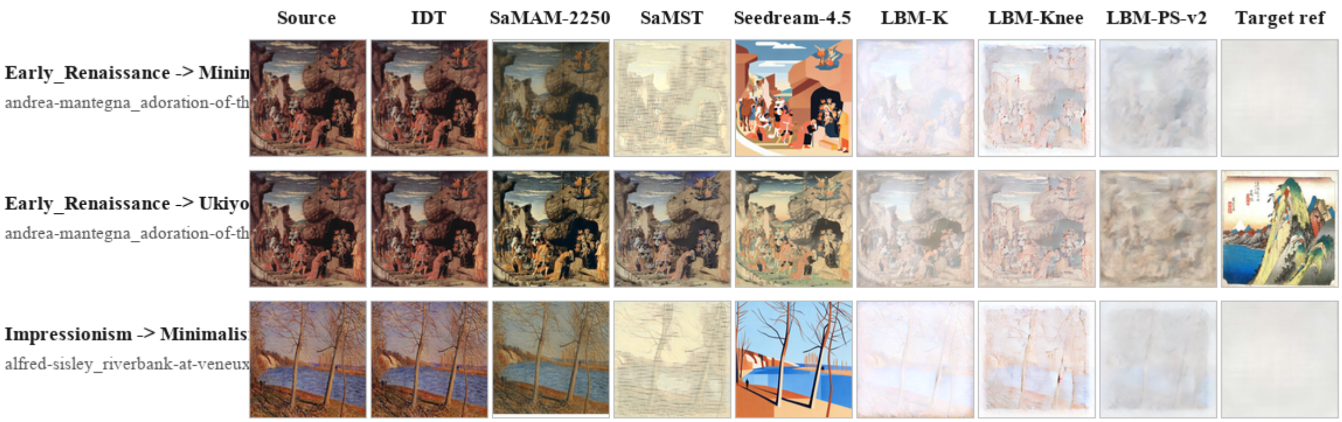

Figure 1: Additional aligned Distinct5 qualitative cases, rows 1–3.

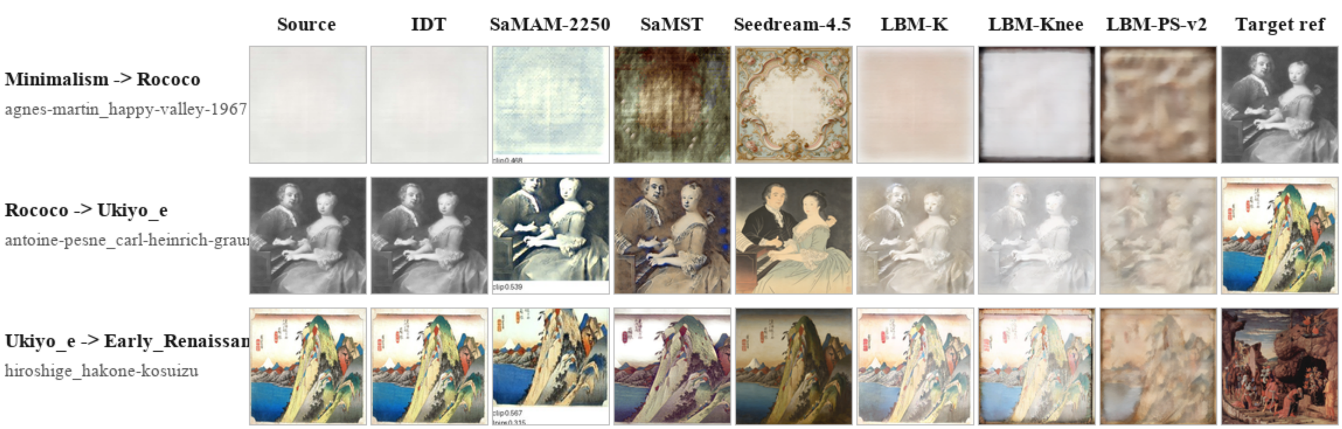

Figure 2: Additional aligned Distinct5 qualitative cases, rows 4–6.